

Aqualytica — AI Water Quality Intelligence Platform

Machine Learning Model Selection & Comparative Evaluation Matrix | Project Defence & Evaluation Review Copy

Model Candidate	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	ROC-AUC	Latency (ms)	Selection Status & Tradeoff Rationale
Random Forest (Selected)	85.80%	93.11%	88.17%	0.906	0.836	0.03 ms	APPROVED — Best accuracy & F1 score; high precision avoids false safety verdicts
Decision Tree (CART)	84.70%	90.27%	89.92%	0.901	0.748	< 0.01 ms	Rejected — High tendency to overfit on marginal sensor voltage fluctuations
Support Vector Classifier (SVC)	81.55%	84.69%	92.95%	0.886	0.823	0.19 ms	Rejected — High computational latency during real-time IoT inference
Gaussian Naive Bayes	81.15%	83.51%	94.25%	0.886	0.823	< 0.01 ms	Rejected — Assumes feature independence; fails for correlated pH and TDS metrics
K-Nearest Neighbors (KNN)	80.75%	83.24%	94.05%	0.883	0.755	0.04 ms	Rejected — High memory footprint during continuous background streaming
Logistic Regression	78.80%	79.76%	97.29%	0.877	0.712	< 0.01 ms	Rejected — Underfitting due to non-linear physical water parameter boundaries

Key Technical Defense Highlights:

- **Random Forest Ensemble Superiority:** Selected as the primary inference engine due to its top-rank Accuracy (85.80%), F1-Score (0.906), and high Precision (93.11%). High precision ensures hazardous water is rarely misclassified as safe.
- **Class Imbalance Mitigation:** Trained using `class_weight="balanced"` and hyperparameter tuned via `RandomizedSearchCV` (150 decision trees, max depth = 15) to prevent majority class prediction bias.
- **Feature Scaling Alignment:** Sensor inputs are normalized via `StandardScaler` prior to inference, matching deployment in the Python Flask microservice.

Project Guide / Supervisor:

External Evaluator Panel: _____

Date: August 5, 2026